

Databases 1

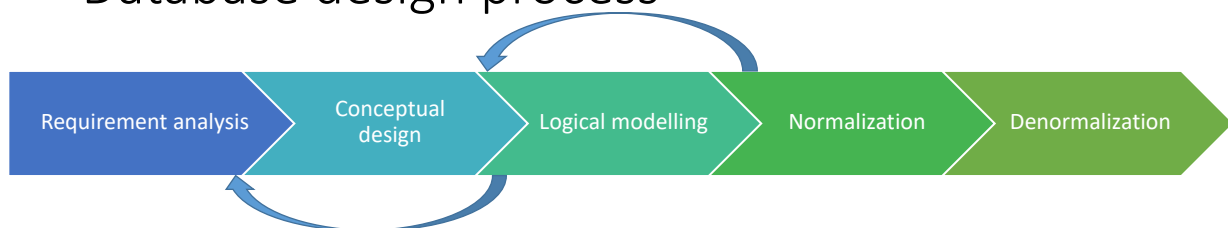
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Lecture 4 – Functional dependencies and normal forms

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Database design process



Conceptual database design gives us:

- a set of **relation schemas**
- **integrity constraints**

Logical constraints:

- functional dependencies
- multivalued dependencies
- join dependencies

Design theory is about how to represent your data to avoid ***anomalies***.

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Anomalies

<u>SID</u>	<u>CID</u>	grade	room
12	CS101	NULL	215WI1
12	CS900	5.0	200WI1
55	CS900	NULL	200WI1
34	CS102	3.5	126WI2

redundancy

<u>SID</u>	<u>CID</u>	grade	room
12	CS101	NULL	215WI1
12	CS900	5.0	200WI1
55	CS900	5.0	013WI1
34	CS102	3.5	126WI2

update anomaly

<u>SID</u>	<u>CID</u>	grade	room
12	CS101	NULL	215WI1
12	CS900	5.0	200WI1
55	CS900	NULL	200WI1
34	CS102	3.5	126WI2
	CS103		224WI2

deletion anomaly

lost information about assigned room

insert anomaly

can't reserve a room without students

<u>SID</u>	<u>CID</u>	grade	room
12	CS900	5.0	200WI1
55	CS900	NULL	200WI1
34	CS102	3.5	126WI2

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Anomalies

Storing the same information redundantly, that is, in more than one place within a database, can lead to several problems:

- **Redundant Storage:** Source information is stored repeatedly.
- **Update Anomalies:** If one copy of such repeated data is updated, an inconsistency is created unless all copies are similarly updated.
- **Insertion Anomalies:** It may not be possible to store certain information unless some other, unrelated, information is stored as well.
- **Deletion Anomalies:** It may not be possible to delete certain information without losing some other, unrelated, information as well.

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Anomalies leads to decomposition

- The essential idea is that many problems arising from redundancy can be addressed by **replacing a relation with a collection of smaller relations**.
- A **decomposition** of a relation schema R consists of replacing the relations schema by two (or more) relation schemas that each contain a subset of the attributes of R and together include all attributes in R .

We must answer two questions:

- Do we need to decompose a relation?
- What problems (if any) does a given decomposition cause?

Normal forms

Two properties of decompositions:

- lossless-join
- dependency-preservation

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Functional dependencies

Definition:

A **functional dependency**, $X \rightarrow Y$, between two sets of attributes X and Y that are subsets of R . The constraint is that, for any two tuples t_1 and t_2 in $r(R)$ that have $t_1[X] = t_2[X]$, they must also have $t_1[Y] = t_2[Y]$.

- The values of the Y component of a tuple in r depend on, or are determined by, the values of the X component; alternatively, the values of the X component of a tuple uniquely (or functionally) determine the values of the Y component.
- We also say that there is a functional dependency from X to Y , or that Y is functionally dependent on X .

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Functional dependencies

$$t_1[X] = t_2[X] \rightarrow t_1[Y] = t_2[Y]$$

- $CID \rightarrow room$
- $\{SID, CID\} \rightarrow \{grade, room\}$



	SID	CID	grade	room
	12	CS101	NULL	215WI1
t_1	12	CS900	5.0	200WI1
t_2	55	CS900	NULL	200WI1
	34	CS102	3.5	126WI2

A functional dependency is a property of the relations schema R , not of a particular legal relation state r of R .

A functional dependency cannot be **inferred automatically** from a given relation extension r but must be defined explicitly by someone who **knows the semantics** of the attributes of R .

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Functional dependencies

- $CID \rightarrow \{cname, ECTS\}$

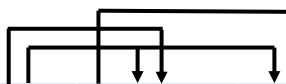
- $CID \rightarrow cname$

- $cname \rightarrow CID$

- $cname \rightarrow ECTS$

- ~~$ECTS \rightarrow cname$~~

- $SID \rightarrow \{first_name, last_name\}$



CID	cname	ECTS
CS101	Programming	5
CS900	Databases	6
CS102	Algorithms	4
EN101	Mechanics	5
EL300	Electronics	6



SID	first_name	last_name
12	Mia	Wallace
34	Vincent	Vega
55	Butch	Coolidge
77	Jules	Winnfield

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Functional dependencies

- Relation extensions $r(R)$ that satisfy the functional dependency constraints are called **legal relation states** of R .
- Hence, the main use of functional dependencies is to describe further a relation schema R by specifying constraints on its attributes that must **hold at all times**.
- Typically, the schema designer specifies the functional dependencies that are semantically obvious; usually, however, numerous other functional dependencies hold in all legal relation instances among sets of attributes that can be derived from and satisfy the dependencies. Those other dependencies can be inferred or deduced from the functional dependencies.

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Normalization

Procedure first proposed by Edgar F. (Ted) Codd:

Codd, E. F. (1972). Further normalization of the data base relational model. Data base systems, 6, 33-64.

1. Perform a conceptual schema design using a conceptual model such as ER and map the conceptual design into a set of relations.
2. Design the relations based on external knowledge derived from an existing implementation of files or forms or reports.
3. The normalization process, takes a relation schema through a series of tests to certify whether it satisfies a certain normal form.

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Normalization

The process by evaluating each relations against the criteria for normal forms and decomposing relations as necessary by finding the **functional dependencies** among the attributes of a relation.

- **first** (1NF) normal form
 - **second** (2NF) normal form
 - **third** (3NF) normal form
- } Initially proposed by Codd
- a stronger definition of 3NF – called **Boyce-Codd** normal form (BCNF)
 - a **fourth** normal form (4NF) – based on concept of **multivalued dependencies**
 - a **fifth** normal form (5NF) – based on concept of **join dependencies**.

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Normalization

The normal form of a relation refers to the **highest normal form** condition that it meets, and hence indicates the degree to which it has been normalized.

Relations must also confirm the existence of properties:

- The nonadditive join **or lossless join property**, which guarantees that the spurious tuple generation does not occur with respect to the relation schemas created after decomposition.
- The **dependency preservation property**, which ensures that each functional dependency is represented in some individual relation resulting after decomposition.

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First Normal Form

- The domain of an attribute must include only **atomic (simple, indivisible) values** and that the value of any attribute in a tuple must be a single value from the domain of that attribute.
- 1NF disallows having a set of values, a tuple of values, or a combination of both as an attribute value for a single tuple.
- 1NF disallows multivalued attributes that are themselves composite (nested relations).

CID	courses
CS101	Programming, 5
CS900	Databases, 6
CS102	Algorithms, 4
EN101	Mechanics, 5

Not in 1NF

courses	students
CS101, Programming	12
CS900, Databases	{12, 55}
CS102, Algorithms	34

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Second Normal Form

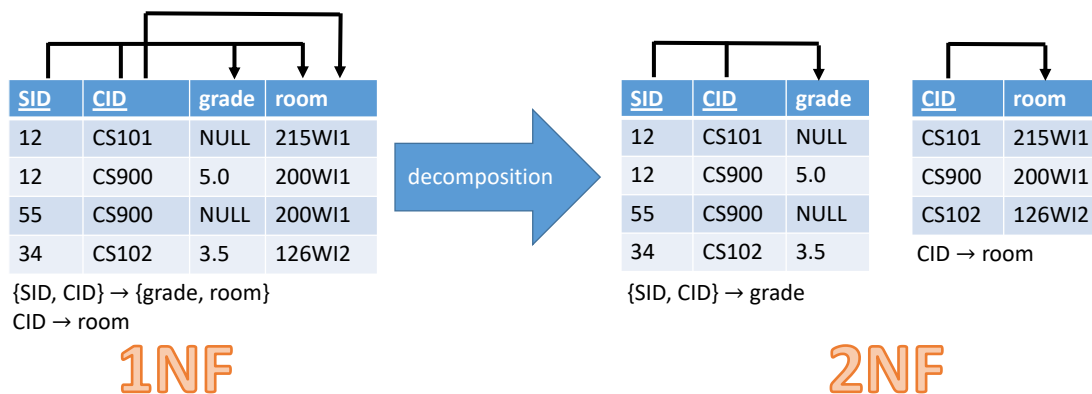
- 2NF is based on the concept of **full functional dependency**.
- A functional dependency $X \rightarrow Y$ is a full functional dependency if removal of any attribute A from X means that the dependency does not hold anymore.
- For any attribute $A \in X$, $(X - \{A\})$ does not functionally determine Y .

A relation schema R is in 2NF if every nonprime attribute A in R is fully functionally dependent on the primary key of R .

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Second Normal Form

A relation schema R is in 2NF if every **nonprime attribute** A in R is **fully functionally dependent** on the primary key of R .



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Third Normal Form

- 3NF is based on the concept of **transitive dependency**.
- A functional dependency $X \rightarrow Y$ in a relation schema R is a transitive dependency if there exists a set of attributes $Z \in R$ that is neither a candidate key nor a subset of any key of R , and both $X \rightarrow Z$ and $Z \rightarrow Y$ hold.

A relation schema R is in 3NF if it satisfies 2NF and no nonprime attribute of R is **transitively dependent** on the primary key.

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Third Normal Form

A relation schema R is in 3NF if it satisfies 2NF and no nonprime attribute of R is **transitively dependent** on the primary key.



<u>SID</u>	<u>CID</u>	grade	room	building
12	CS101	NULL	215W11	W11
12	CS900	5.0	200W11	W11
55	CS900	NULL	200W11	W11
34	CS102	3.5	126W12	W12

$\{SID, CID\} \rightarrow \{grade, room, building\}$
 $CID \rightarrow room$
 $\{SID, CID\} \rightarrow \{grade, room\}$
 $room \rightarrow building$

decomposition



<u>SID</u>	<u>CID</u>	grade
12	CS101	NULL
12	CS900	5.0
55	CS900	NULL
34	CS102	3.5

$\{SID, CID\} \rightarrow grade$



<u>CID</u>	<u>room</u>	building
CS101	215W11	W11
CS900	200W11	W11
CS102	126W12	W12

$room \rightarrow building$

$CID \rightarrow room$



<u>CID</u>	<u>room</u>	<u>building</u>
CS101	215W11	W11
CS900	200W11	W11
CS102	126W12	W12

$CID \rightarrow room$
 $room \rightarrow building$
 $CID \rightarrow \{room, building\}$

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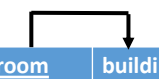
Third Normal Form

A relation schema R is in 3NF if it satisfies 2NF and no nonprime attribute of R is **transitively dependent** on the primary key.



<u>SID</u>	<u>CID</u>	grade
12	CS101	NULL
12	CS900	5.0
55	CS900	NULL
34	CS102	3.5

$\{SID, CID\} \rightarrow grade$



<u>room</u>	<u>building</u>
215W11	W11
200W11	W11
126W12	W12

$room \rightarrow building$



<u>CID</u>	<u>room</u>
CS101	215W11
CS900	200W11
CS102	126W12

$CID \rightarrow room$

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Normalization action

Resulting normal form	Normalization action
1NF	Form new relations for each multivalued attribute or nested relation.
2NF	Decompose and set up a new relations for each partial key with its dependent attribute(s). Make sure to keep a relations with the original primary key and any attributes that are fully functionally dependent on it.
3NF	Decompose and set up a relation that includes the nonkey attribute(s) that functionally determine(s) other nonkey attribute(s).

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Boyce-Codd normal form (BCNF)

A relation schema R is in BCNF if whenever a **nontrivial functional dependency** $X \rightarrow Y$ holds in R , then X is a superkey of R .

- **Nontrivial functional dependency:** If functional dependency $X \rightarrow Y$ holds, then $X \cap Y = \emptyset$ or in other words $X \not\supseteq Y$.
- Every relation in BCNF is also in 3NF; a relation in 3NF is not necessarily in BCNF.
- Only in rare cases does a 3NF relation not meet the requirements of BCNF.
- A 3NF relation that does not have multiple overlapping candidate keys is guaranteed to be in BCNF.

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Boyce-Codd normal form (BCNF)

A relation schema R is in BCNF if whenever a **nontrivial functional dependency** $X \rightarrow Y$ holds in R , then X is a superkey of R .

A 3NF relation won't be in BCNF if:

- there are multiple candidate keys,
- the keys are composed of multiple attributes,
- there are common attributes between the keys.

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Boyce-Codd normal form (BCNF)

A relation schema R is in BCNF if whenever a **nontrivial functional dependency** $X \rightarrow Y$ holds in R , then X is a superkey of R .

Lecturer	Date	Room
Daenerys Targaryen	2022-03-01 10:00	010
Jon Snow	2022-03-01 12:00	215
Cersei Lannister	2022-03-02 14:00	200
Lord Varys	2022-02-28 9:00	128

Assumptions:

- Only one lecturer per room
- Lecturer newer move

3NF

Superkeys:

- {Lecturer, Date, Room}
- {Lecturer, Date} **Candidate key**
- {Date, Room} **Candidate key**

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Decomposition of Relations not in BCNF

Lecturer	Date	Room
Daenerys Targaryen	2022-03-01 10:00	010
Jon Snow	2022-03-01 12:00	215
Cersei Lannister	2022-03-02 14:00	200
Lord Varys	2022-02-28 9:00	128

• FD1: {Lecturer, Date} → Room

• FD2: Lecturer → Room

Could be decomposed into:

R1: (Lecturer, Date), R2: (Date, Room)

R1: (Lecturer, Room), R2: (Date, Room)

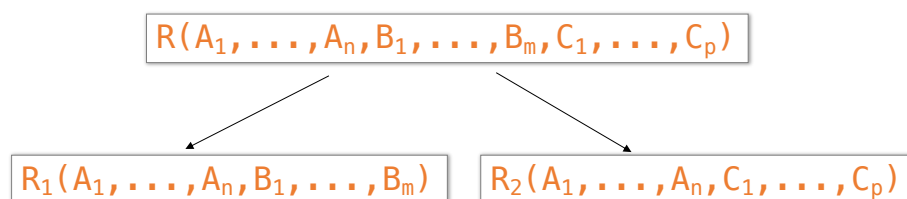
R1: (Lecturer, Room), R2: (Lecturer, Date)

Properties of decomposition to meet during normalization proces:

1. the nonadditive join
2. the functional dependency preservation

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Lossless Decompositions

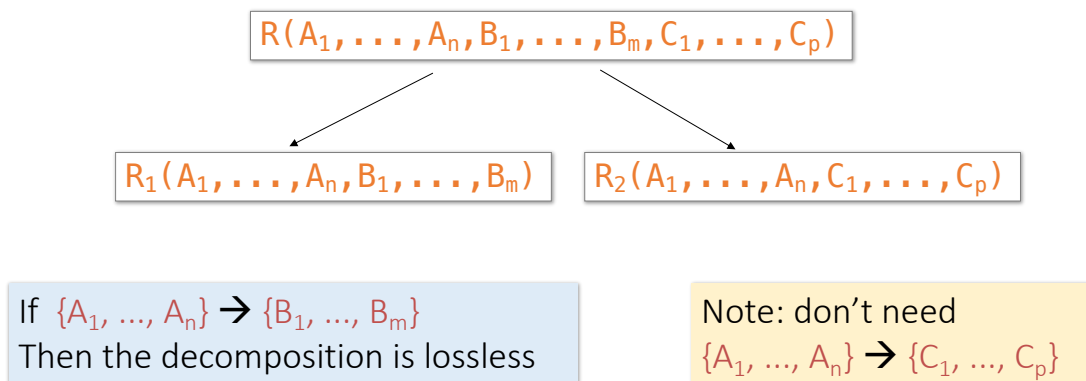


A decomposition R to (R1, R2) is lossless if:

1. $R = R1 \cup R2$
2. $R1 \cap R2$ is a superkey for R1 or R2

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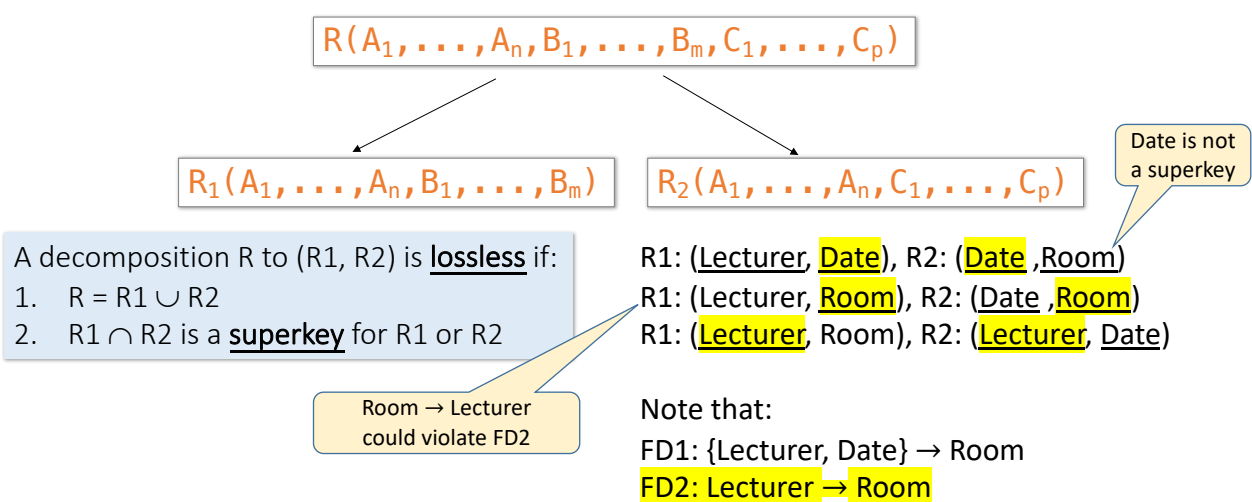
Lossless Decompositions



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Lossless Decompositions



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Boyce-Codd normal form (BCNF)

A relation schema R is in BCNF if whenever a **nontrivial functional dependency** $X \rightarrow Y$ holds in R , then X is a superkey of R .

<u>Lecturer</u>	<u>Date</u>	Room
Daenerys Targaryen	2022-03-01 10:00	010
Jon Snow	2022-03-01 12:00	215
Cersei Lannister	2022-03-02 14:00	200
Lord Varys	2022-02-28 9:00	128



<u>Lecturer</u>	Room
Daenerys Targaryen	010
Jon Snow	215
Cersei Lannister	200
Lord Varys	128

<u>Lecturer</u>	<u>Date</u>
Daenerys Targaryen	2022-03-01 10:00
Jon Snow	2022-03-01 12:00
Cersei Lannister	2022-03-02 14:00
Lord Varys	2022-02-28 9:00

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Multivalued Dependency

- Some relations have constraints that cannot be specified as functional dependencies and hence are not in violation of BCNF.

A **multivalued dependency** $X \twoheadrightarrow Y$ specified on relation schema R , where X and Y are both subsets of R , specifies the following constraint on any relation state r of R : If two tuples t_1 and t_2 exist in r such that $t_1[X] = t_2[X]$, then two tuples t_3 and t_4 should also exist in r with the following properties,

where $Z = (R - (X \cup Y))$:

- $t_1[X] = t_2[X] = t_3[X] = t_4[X]$
- $t_1[Y] = t_3[Y]$ and $t_2[Y] = t_4[Y]$
- $t_2[Z] = t_3[Z]$ and $t_1[Z] = t_4[Z]$

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Multivalued Dependency

Theatre	Movie title	Language
1	Pulp Fiction	English
1	The Shawshank Redemption	Polish
1	Pulp Fiction	Polish
1	The Shawshank Redemption	English
2	The Godfather	English
2	The Godfather	Polish

- $t_1[X] = t_2[X] = t_3[X] = t_4[X]$
- $t_1[Y] = t_3[Y]$ and $t_2[Y] = t_4[Y]$
- $t_2[Z] = t_3[Z]$ and $t_1[Z] = t_4[Z]$

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Fourth Normal Form

A relation schema R is in 4NF if whenever:

1. $X \twoheadrightarrow Y$ is a **nontrivial multivalued dependencies**
2. X is a superkey.

- An **all-key relation** is always in BCNF since it has no FDs.
- An all-key relation, which has no FDs but has the MVD, is not in 4NF.
- A relation that is not in 4NF due to a nontrivial MVD must be decomposed to convert it into a set of relations in 4NF.
- The decomposition removes the redundancy caused by the MVD.

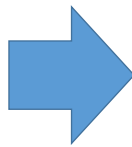
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Forth Normal Form

A relation schema R is in 4NF if whenever:

1. $X \twoheadrightarrow Y$ is a **nontrivial multivalued dependencies**
2. X is a superkey.

<u>Theatre</u>	<u>Movie_title</u>	<u>Language</u>
1	Pulp Fiction	English
1	Pulp Fiction	Polish
1	The Shawshank Redemption	Polish
1	The Shawshank Redemption	English
2	The Godfather	English
2	The Godfather	Polish



<u>Theatre</u>	<u>Movie_title</u>
1	Pulp Fiction
1	The Shawshank Redemption
2	The Godfather

<u>Theatre</u>	<u>Language</u>
1	English
1	Polish
2	English
2	Polish

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Summary - design guidelines

1. Do not combine attributes from multiple entity types and relationship types into a single relation.
2. Design the base relation schemas so that no insertion, deletion, or modification anomalies are present in the relations. If any anomalies are present, note them clearly and make sure that the programs that update the database will operate correctly.
3. As far as possible, avoid placing attributes in a base relation whose values may frequently be NULL. If NULLs are unavoidable, make sure that they apply in exceptional cases only and do not apply to a majority of tuples in the relation.
4. Design relation schemas so that they can be joined with equality conditions on attributes that are appropriately related (primary key, foreign key) pairs in a way that guarantees that no artificial tuples are generated.

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Denormalization – controlled redundancy

- A completely normalized database may not be the most efficient or effective implementation.
- To retrieve complete information from a normalized database, the JOIN operation must be used frequently.
- JOIN tends to be expensive in terms of processing time, and very large joins on many tables are very expensive.

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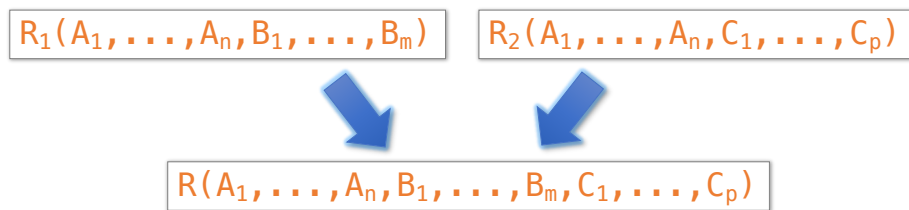
Denormalization strategies

1. Collapsing Tables
2. Splitting Tables (Horizontal/Vertical Splitting)
3. Removing dictionaries
4. Derived Attributes (Summary, Total, and Balance)

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Denormalization strategies

Collapsing Tables

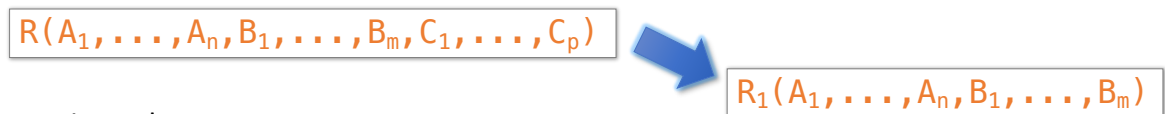


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Denormalization strategies

Splitting Tables

- Vertical



- Horizontal

Orders

OrderID	CustomerID	Date
134532	65675433	2025-06-22
574331	87658845	2025-09-30
767531	34676453	2024-01-02
943521	25567324	2024-12-31



Orders_2025

OrderID	CustomerID	Date
134532	65675433	2025-06-22
574331	87658845	2025-09-30

Orders_2024

OrderID	CustomerID	Date
767531	34676453	2024-01-02
943521	25567324	2024-12-31

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Denormalization strategies

Removing dictionaries

<u>EmpID</u>	Name	CountryID
E0045	Smith	4567
E3542	Mike	4567
E1111	Smith	1234
E9999	Mary	1234

<u>CountryID</u>	Country
1234	Canada
4567	United States



<u>EmpID</u>	Name	Country
E0045	Smith	United States
E3542	Mike	United States
E1111	Smith	Canada
E9999	Mary	Canada

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Denormalization

Derived Attributes

```
Order(orderID, date, customerID, deliveryID)
```

```
OrderItem(itemID, orderID, itemPrice, quantity)
```



```
Order(orderID, date, customerID, deliveryID, orderPrice, numberOfItems)
```

```
OrderItem(itemID, orderID, itemPrice, quantity)
```

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Surrogate key

- An artificial simple key used to identify individual entities when there is no natural key or when the situation demands a noncomposite key, but no natural noncomposite key exists.
- Surrogate keys do not identify individual entities in a meaningful way: they are simply an arbitrary method to distinguish among them.
- The surrogate is internally generated by the system but is nevertheless visible to the user or application.
- UUID - Universally unique identifier – 32 hexadecimal digits

123e4567-e89b-12d3-a456-426652340000

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Surrogate key

Intelligent attribute -
has some meaning that
we do not want to leak

ID_number	Lecturer	Room
03323154326	Daenerys Targaryen	010
80010112345	Jon Snow	215
2222222222	Cersei Lannister	200
98030112345	Lord Varys	128

Surrogate key

UUID	ID_number	Lecturer	Room
f9b18268-995f-11ec-b909-0242ac120002	03323154326	Daenerys Targaryen	010
f9b18506-995f-11ec-b909-0242ac120002	80010112345	Jon Snow	215
f9b18664-995f-11ec-b909-0242ac120002	2222222222	Cersei Lannister	200
f9b18790-995f-11ec-b909-0242ac120002	98030112345	Lord Varys	128

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Thank you for your
attention!

Time for questions

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